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PAPER_1086: SCm Dark Energy Density with Γ-Coupled Phonon Modulation Replacing ΛCDM

Abstract

We derive the Γ-coupled SCm dark energy density:

$$\rho_{\mathrm{DE}}(t, \Gamma) = \rho_{\mathrm{SCm}}(t) \cdot S_{26} \cdot \Phi(\Gamma) \cdot (2R - 1)$$

This replaces the static ΛCDM cosmological constant $\rho_\Lambda = 0.692\rho_{\mathrm{crit}}$ with a dynamical dark energy density sourced by the SCm vacuum, modulated by phonon resonance at 1.25 THz. The ratio $\rho_{\mathrm{DE}}/\rho_\Lambda$ quantifies the departure from ΛCDM. Distinct from PAPER_1076 (which derives $w(z)$ from linewidth broadening) and PAPER_889 (ΛCDM contrast), this paper provides the explicit Γ-coupled density computation.

§1 SCm Vacuum Density Evolution

$$\rho_{\mathrm{SCm}}(t) = \rho_{\mathrm{vac,SCm}} \cdot S_{26} \cdot \exp\left(\kappa t + \frac{[\mathrm{SSq}]}{26} t\right)$$

with $\rho_{\mathrm{vac,SCm}} = 9.47 \times 10^{-27}$ kg/m^3, $\kappa = 5.787 \times 10^{-9}$ s^{-1}.

§2 Gamma-Coupled Dark Energy Density

$$\rho_{\mathrm{DE}}(t, \Gamma) = \rho_{\mathrm{SCm}}(t) \cdot S_{26} \cdot \Phi(\Gamma) \cdot (2R - 1)$$

where $R = F_{U,Bi}/F_U$ determines the expansion/erosion sign.

§3 Comparison to ΛCDM

$$\rho_\Lambda = 0.692 \cdot \rho_{\mathrm{crit}} = 0.692 \cdot \frac{3H_0^2}{8\pi G}$$

$$\frac{\rho_{\mathrm{DE}}}{\rho_\Lambda} = \frac{\rho_{\mathrm{SCm}}(t) \cdot S_{26} \cdot \Phi \cdot (2R - 1)}{0.692\rho_{\mathrm{crit}}}$$

At $t = 0$, $R = 0.8$: $\rho_{\mathrm{DE}}/\rho_\Lambda \sim 10^{22}$ (phonon amplification dominates at resonance).

§4 Physical Advantages over ΛCDM

Aspect	ΛCDM	SCm Dark Energy
Origin	Ad-hoc constant Λ	SCm vacuum buoyancy
Time evolution	Static $w = -1$	Dynamic via $\rho_{\mathrm{SCm}}(t)$
Free parameters	Ω_Λ (fine-tuned)	κ, [SSq] (calibrated)
Phonon coupling	None	$\Phi_{1.25\,\mathrm{THz}}(\Gamma)$
Fine-tuning problem	10^{120} discrepancy	2-parameter resolution
Testability	Indirect (supernovae)	Direct (THz phonon spectrum)

§5 Linewidth Sweep

Γ (THz)	Φ	ρ_{DE} (kg/m^3)	$\rho_{\mathrm{DE}}/\rho_\Lambda$
0.01	1.45×10^{21}	2.18×10^{-4}	2.3×10^{22}
0.10	1.45×10^{21}	2.18×10^{-4}	2.3×10^{22}
1.00	1.45×10^{21}	2.18×10^{-4}	2.3×10^{22}

At resonance, Φ is independent of Γ (Gaussian peak = 1), so off-resonance behavior drives the Γ-dependence.

References

- PAPER_1076: SCm Dark Energy with Phonon Linewidth Γ-Modulation
- PAPER_889: EtVsLambdaCDMDarkEnergyContrastCalc
- PAPER_890: SCmVacuumDensityEvolutionCalc
- PAPER_1087: Dark Energy Equation of State w_DE

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